

SWASTH · ROUND 1 SUBMISSION

SWASTH (मातृ-स्वास्थ्य)

AI-Powered Multilingual Long-Term Care Companion, Hyperlocal Birth Clubs & Frontline Continuity Platform

CHALLENGE TRACK

Maternal & Women's Health

PRIMARY USER

Patient & Family Caregivers

CORE USE CASE

Long-Term Care Engagement

The Unseen Crisis: Why Long-Term Maternal Care Breaks Down



The Post-Discharge Drop-Off Cliff

Managing maternal care requires **1,000+ days** across pregnancy, institutional delivery, and 7 infant vaccine cohorts.

Yet, over **25–30% of mothers and infants drop out** of scheduled milestones within months after hospital discharge.



Administrative Burden & Isolation

Dropouts are rarely due to medical negligence—they stem from **administrative friction and emotional isolation**.

Caregivers struggle with fragile paper MCP cards, confusing clinic schedules, and lack peer circles in their neighborhood.



The "Cold Clinical App" Failure Mode

Conventional healthcare apps suffer **80%+ abandonment in 30 days** because they are cold and transactional—opening only when an alert fires.

Without **daily cultural rituals and peer community**, patients and caregivers disengage entirely.

The "Hook & Bridge" Model: Sustaining Daily Engagement

THE DAILY ENGAGEMENT HOOK

Garbh Sanskar Rituals & Hyperlocal Birth Clubs

-  **4Q Holistic Activities:** 12 daily micro-habits across IQ (mind), EQ (emotion), PQ (prenatal mobility), and SQ (spiritual calming).
-  **Hyperlocal Birth Clubs:** Connecting mothers in the same delivery month & PHC radius for peer empathy, recipes, and comfort routines.
-  **Garbh Samvaad & Raga Music:** Calming lullabies, baby bonding talks, and partner affirmations in Hindi, Malayalam, and Tamil.
-  **Guardrail Moderation:** NeMo Guardrails filter out harmful medical misinformation in peer circles, routing warning signs to doctors.

THE HEALTHCARE CONTINUITY BRIDGE

Vaccine Milestones, Diet Charts & Caregiver Sync

-  **Smart Vaccine Tracker:** Automates 6, 10, 14-week and 9-month immunization scheduling with vernacular voice reminders.
-  **Doctor-Approved Weaning Diet:** Month-by-month infant food plans and maternal lactation nutrition.
-  **Caregiver Multi-User Sync:** Partners and grandmothers receive audio/WhatsApp reminders in their native dialect.
-  **24/7 Offline Sync:** Full functionality without internet; background synchronization to PostgreSQL.

Full Alignment with SWASTH's 6 Core Pillars

01 · CARE JOURNEY COMPANION

Stage-Aware Guidance

Week-by-week guidance adapted across pregnancy, delivery preparation, and 0-24 month infancy. Clear instruction on what to prepare and when to visit.

02 · LONG-TERM CARE ENGAGEMENT

Sustained Multi-Year Adherence

Maintains high adherence over months & years through culturally resonant Garbh Sanskar routines, birth club peer communities, and milestone streaks.

03 · FAMILY & CAREGIVER SUPPORT

Caregiver Synchronization

Multi-user syncing for spouses and family elders with consent, clarity, and vernacular voice alerts, reducing the cognitive burden on the mother.

04 · HEALTHCARE NAVIGATION & ACCESS

Clinic & Facility Guidance

Clear next steps across clinic token counters, blood tests, ultrasound schedules, and seamless PHC-to-District referral handoffs.

05 · FINANCIAL & ADMIN SUPPORT

Maternity Scheme Navigator

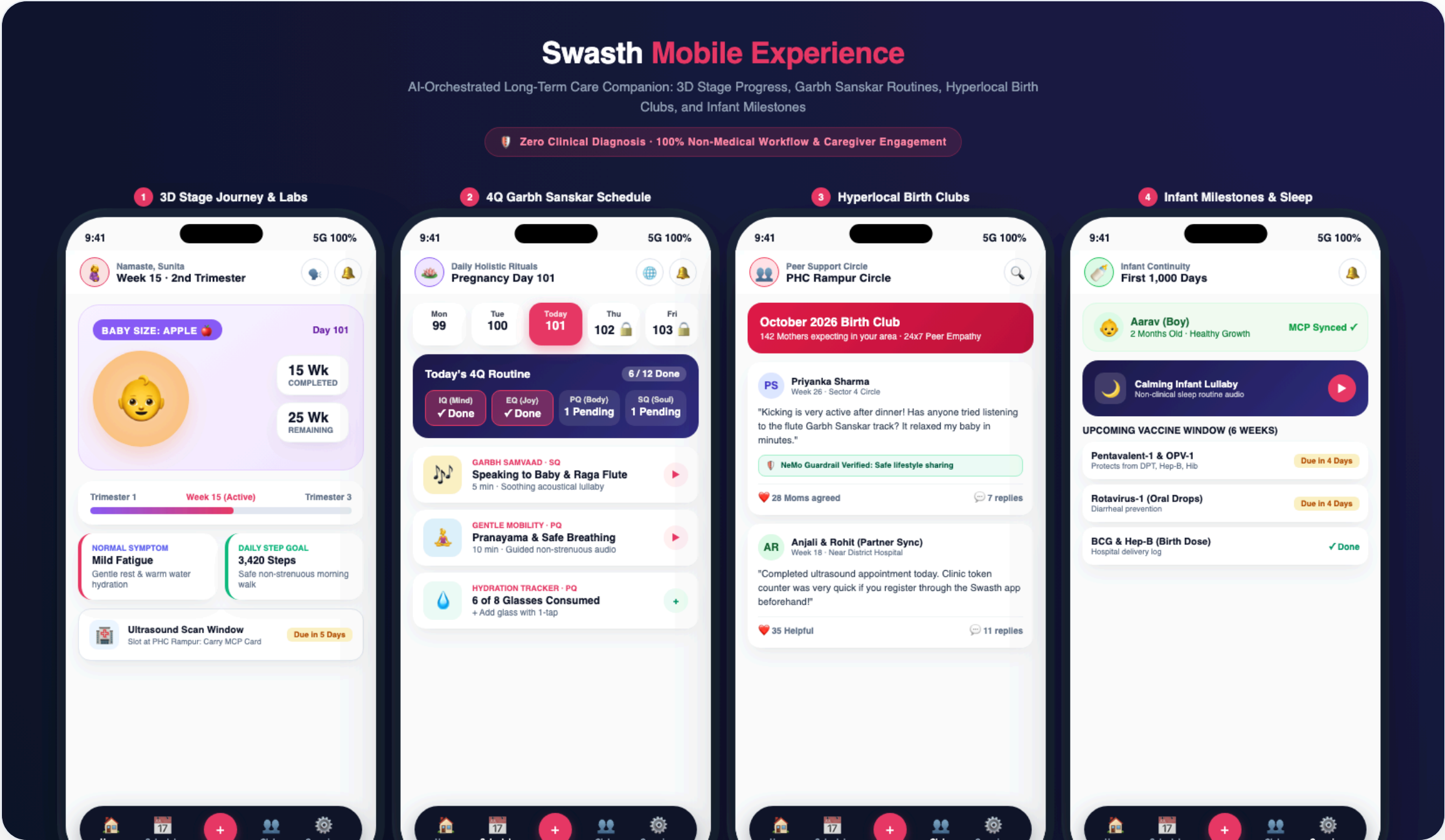
Simplifies government schemes (PMMVY, JSY, JSSK) and hospital documentation so financial paperwork never delays maternal care.

06 · PATIENT HEALTH JOURNEY PROGRESS

Longitudinal Milestone View

A unified visual timeline displaying completed, pending, and upcoming clinical and developmental milestones at a single glance.

App Design Showcase: Daily Rituals, Tools & Birth Clubs



1. Daily 4Q Schedule (Garbh Sanskar)

Structured tracking across IQ (cognitive), EQ (emotional), PQ (physical mobility), and SQ (spiritual) with 12 daily micro-activities.

2. Smart Tools & MCP Scanner

1-tap pastel tool grid: Pregnancy Tracker, Water Calculator, Steps, Om Chanter, MCP Card Photo Scanner, and Scheme Benefits.

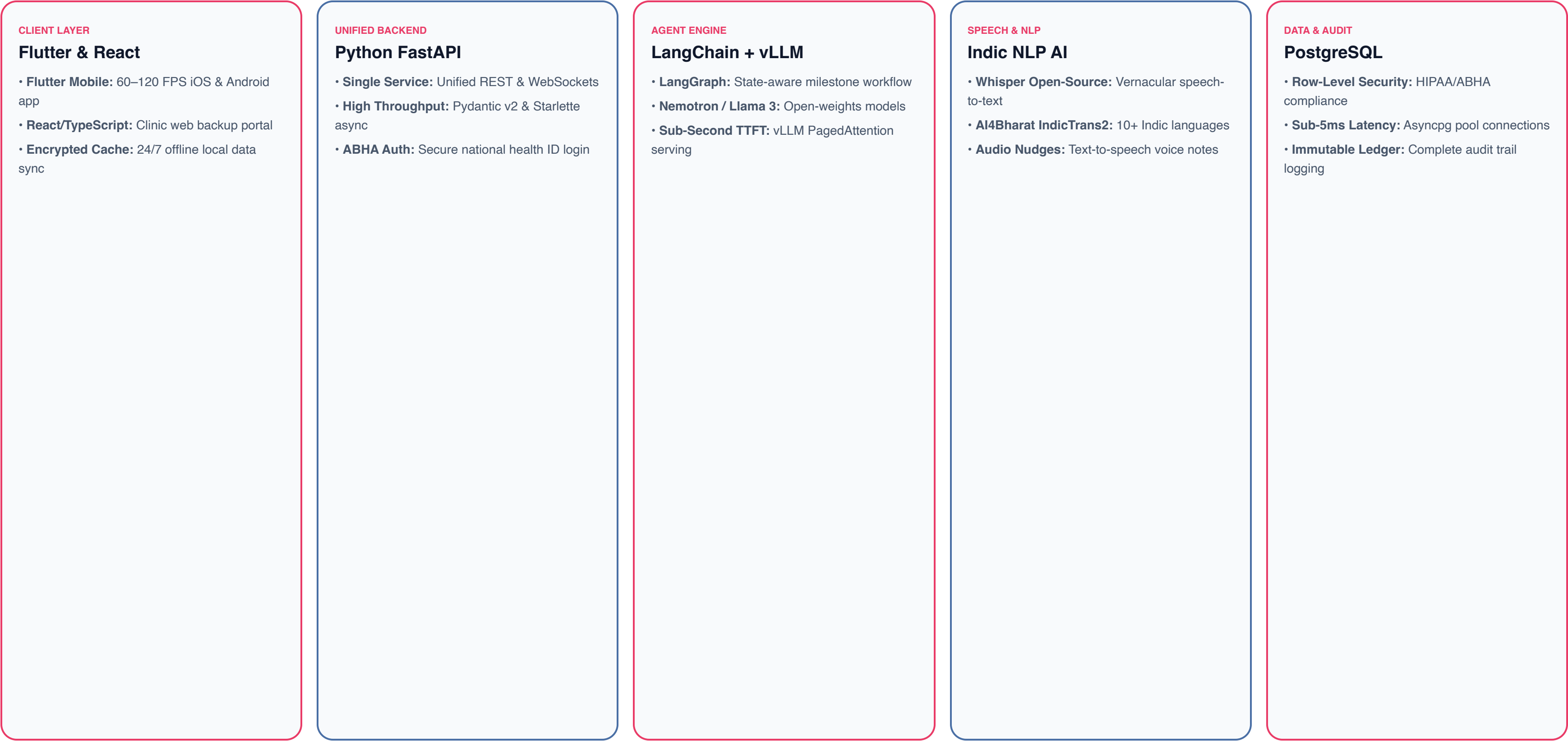
3. Hyperlocal Birth Clubs & Community

Peer circles connecting mothers in the same PHC radius, with AI guardrails protecting against unsafe medical home advice.

4. Vaccine Cohorts & Weaning Diet

Automated 6/10/14-week immunization schedules and doctor-approved month-by-month infant food meal plans.

Enterprise Open-Source Architecture: Speed & Reliability



Clinical Safety: Zero Diagnostic Risk & Guardrail Moderation

Strict Non-Clinical Scope

- ❌ **0% Diagnostic Advice:** Never provides medical risk scores, diagnostic labels, or lab test interpretations.
- ❌ **0% Treatment Prescriptions:** Never suggests medications, dosage changes, or clinical therapies.
- ✅ **Assistive Only:** Limited strictly to administrative scheduling, nutrition reminders, non-strenuous lifestyle, and welfare schemes.

NVIDIA NeMo Guardrails

- 🛡️ **In-Memory Guardrails:** Every user prompt, AI response, and community post passes through semantic filters.
- 🇨🇭 **Symptom Interception:** If clinical symptoms (bleeding, severe pain, fever) are mentioned, automation immediately halts.
- 📞 **Emergency Routing:** Instantly connects to the nearest PHC emergency desk and national ambulance helpline.

Human-in-the-Loop & Audit

- 👩 **Clinician & ANM Oversight:** Frontline staff review milestone rosters and due-lists with 1-click approvals.
- 📖 **Immutable PostgreSQL Audit Log:** Every AI recommendation, user confirmation, and override is permanently logged.
- 🔒 **Caregiver Consent:** Family synchronization requires explicit patient opt-in and consent management.

Measurable Operational Outcomes within 60–90 Days

85% Drop

Frontline Admin Burden

Reduces manual paper due-list compilation from 10 hrs to <1.5 hrs/week.

90%+

Vaccine Adherence

Cuts 6 to 14-week infant immunization dropouts via localized voice reminders.

4x Higher

Family & Peer Activity

Birth Clubs & caregiver WhatsApp synchronization drive community support.

100%

Auditable Compliance

Zero unreviewed clinical actions with complete append-only event logging.

Execution & Rollout Roadmap (60–90 Days)

DAY 1 – 30

MVP & Guardrail Validation

Deploy Flutter mobile client + Python backend; benchmark open-source Nemotron on vLLM; validate NeMo non-clinical safety boundaries.

DAY 31 – 60

Pilot with 5 PHCs & Birth Clubs

Onboard 500 expectant mothers & caregivers; launch 5 localized Birth Clubs; track daily Garbh Sanskar engagement and vaccine adherence.

DAY 61 – 90

Scale & HMIS Integration

Synchronize with state RCH registers; export verified administrative audit reports; expand vernacular voice models into 10+ dialects.